

LOW-COST PULSE DRIVER CIRCUIT For Solenoids And Coils

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A pulse driver circuit designed for solenoids and coils generates a short pulse of predetermined width to activate the solenoid, regardless of the duration of the input drive signal. This compact solenoid driver is useful for actuating solenoids in various devices and models, such as pinball machines or coil guns.

Fig. 1 shows a pinball coil lead, while Fig. 2 illustrates a drive pulse used in a pinball coil lead.

For details, refer to the links: <https://en.wikipedia.org/wiki/Pinball> for the pinball game and <https://en.wikipedia.org/wiki/Coilgun> for coil gun.

Circuit diagram and working

Fig. 3 presents the circuit diagram of the solenoid pulse driver. It is built around 5V voltage regulator LM7805 (IC1), CD4093 (IC2), optocoupler PC817 (IC3), MOSFET IRLZ44 (T1), and a 12V DC solenoid. The design is simple and self-explanatory, utilising standard budget components.



Fig. 1: A pinball coil lead

IC1 provides a regulated 5V output to power the circuit. The optocoupler IC3 sends a pulse to the NAND gates. The NAND gate monostable circuit, built around CD4093 (IC2), activates the solenoid.

As the logic gates do not supply enough current to drive the solenoid, logic-level MOSFET

IRLZ44 (T1) is employed as the power driver. The 100Ω resistor (R4) limits the gate current and prevents ringing, while the $10k\Omega$ resistor (R5) acts as a pull-down resistor to ensure MOSFET T1 remains in a known state.

A solenoid is a coil that generates a magnetic field when current

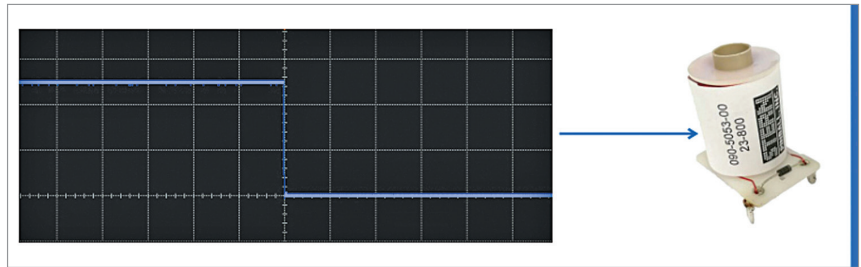


Fig. 2: Drive pulse

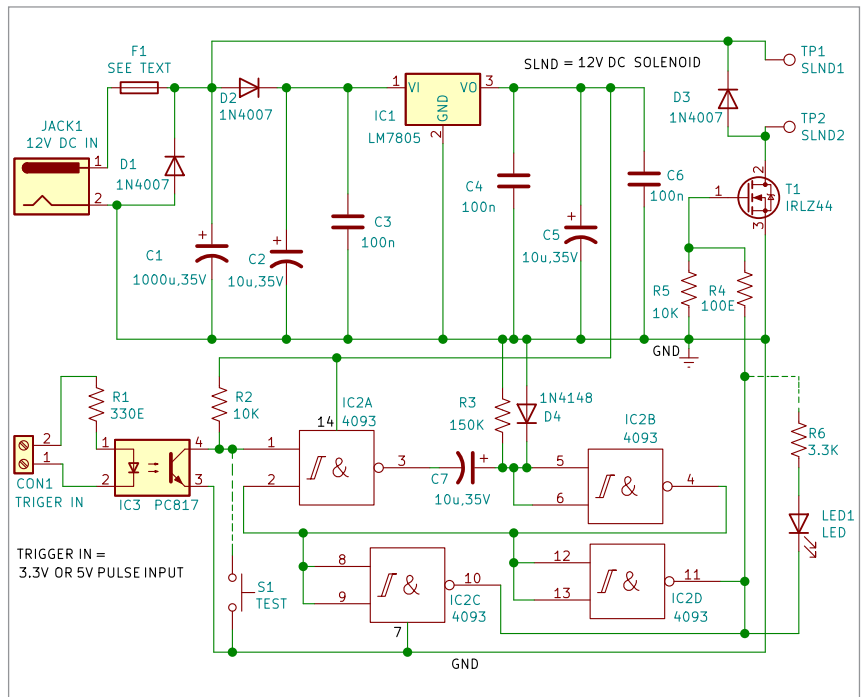


Fig. 3: Circuit diagram